

DOS Project

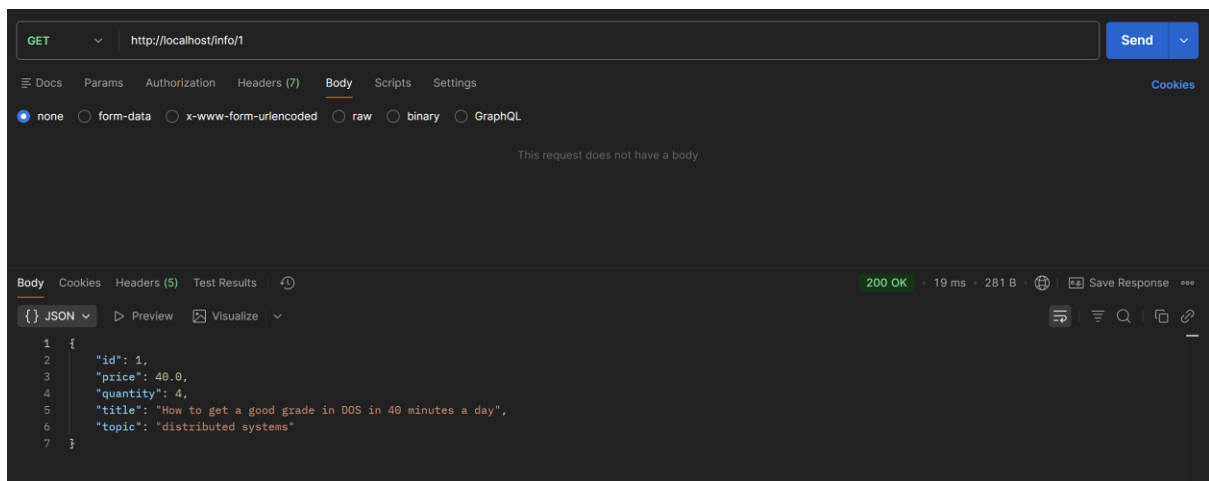
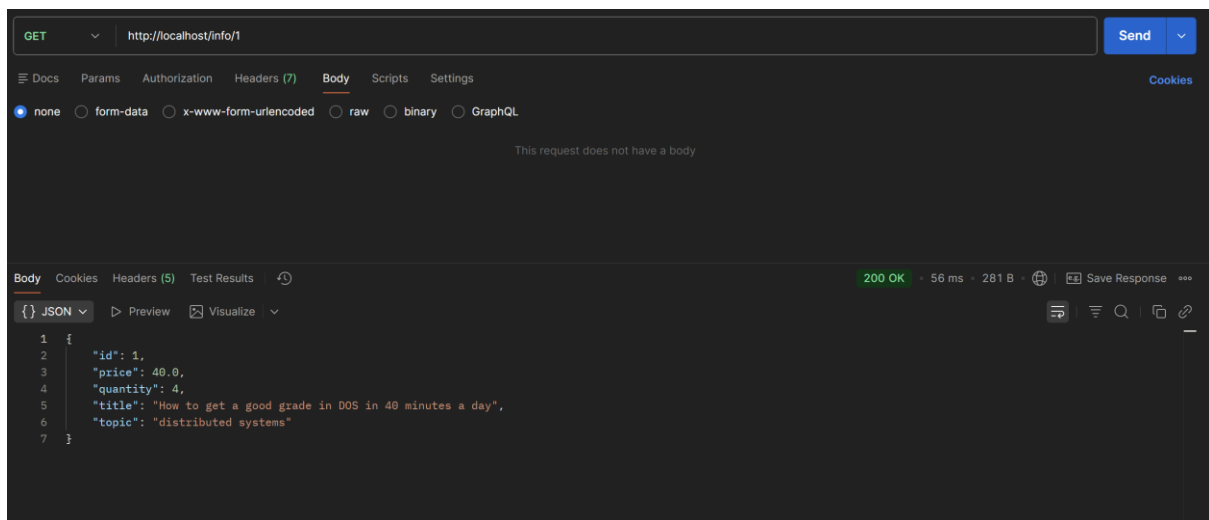
Part 2

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No cache

When caching is disabled, all info requests result in CACHE MISS, and every request is forwarded to the catalog replicas.



```
frontend | 2025-12-22 18:50:53,520 [INFO] 172.19.0.7 - - [22/Dec/2025 18:50:53] "GET /info/1 HTTP/1.0" 200 -
frontend | 2025-12-22 18:50:54,503 [INFO] CACHE MISS info:1
frontend | 2025-12-22 18:50:54,510 [INFO] 172.19.0.7 - - [22/Dec/2025 18:50:54] "GET /info/1 HTTP/1.0" 200 -
frontend | 2025-12-22 18:50:55,433 [INFO] CACHE MISS info:1
frontend | 2025-12-22 18:50:55,442 [INFO] 172.19.0.7 - - [22/Dec/2025 18:50:55] "GET /info/1 HTTP/1.0" 200 -
```

GET

http://localhost/search/undergraduate

Send

Docs

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

200 OK

18 ms

331 B

Save Response

{ } JSON

Preview

Visualize

```
1  [  
2    {  
3      "id": 3,  
4      "title": "Xen and the Art of Surviving Undergraduate School"  
5    },  
6    {  
7      "id": 4,  
8      "title": "Cooking for the Impatient Undergrad"  
9    },  
10   {  
11     "id": 6,  
12     "title": "Why theory classes are so hard"  
13   }  
14 ]
```

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Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

200 OK

11 ms

331 B

Save Response

{ } JSON

Preview

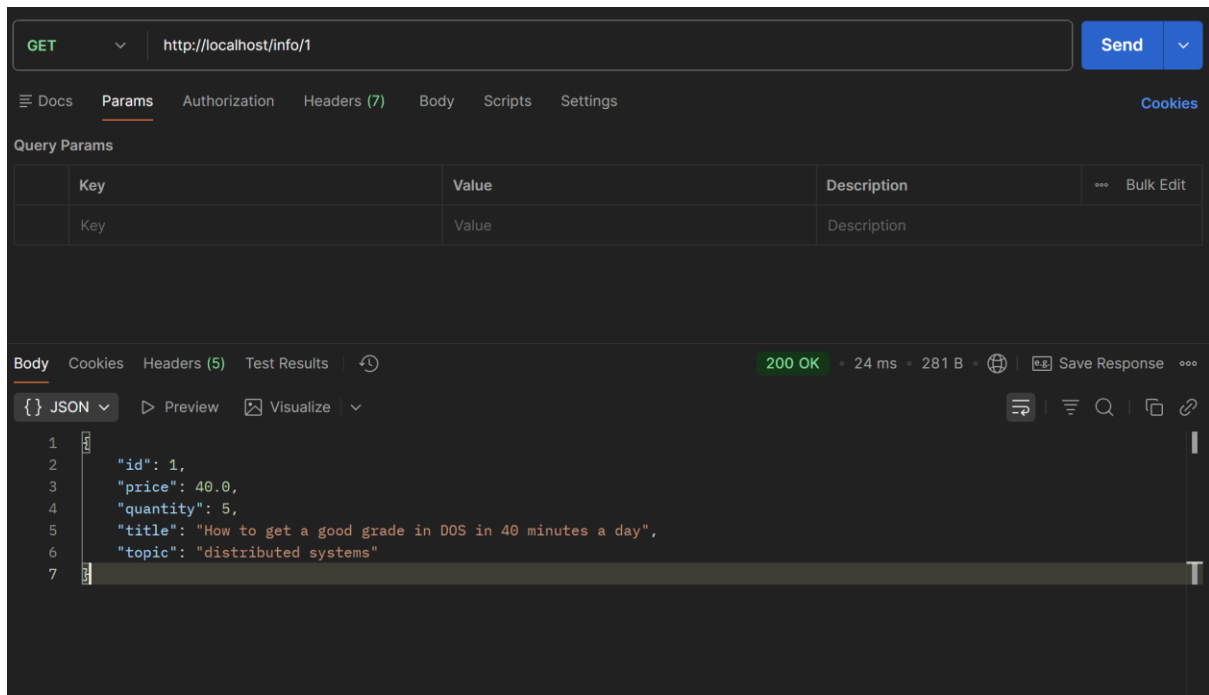
Visualize

```
1  [  
2    {  
3      "id": 3,  
4      "title": "Xen and the Art of Surviving Undergraduate School"  
5    },  
6    {  
7      "id": 4,  
8      "title": "Cooking for the Impatient Undergrad"  
9    },  
10   {  
11     "id": 6,  
12     "title": "Why theory classes are so hard"  
13   }  
14 ]
```

With cache

The first info request results in a CACHE MISS.

The subsequent request for the same item results in a CACHE HIT, confirming that the response is served from the in-memory cache.



GET http://localhost/info/1

Send

Docs Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (5) Test Results

200 OK • 24 ms • 281 B • Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "price": 40.0,
4   "quantity": 5,
5   "title": "How to get a good grade in DOS in 40 minutes a day",
6   "topic": "distributed systems"
7 }
```

```
PS C:\Users\asus\dos project part 1\bazar-online-book-store\bazar-online-book-store-main> docker compose logs -f frontend
frontend | * Serving Flask app 'app'
frontend | * Debug mode: off
frontend | 2025-12-21 21:14:21,497 [INFO] WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
frontend | * Running on all addresses (0.0.0.0)
frontend | * Running on http://127.0.0.1:5000
frontend | * Running on http://172.18.0.6:5000
frontend | 2025-12-21 21:14:21,497 [INFO] Press CTRL+C to quit
frontend | 2025-12-21 21:15:02,937 [INFO] CACHE MISS info:1
frontend | 2025-12-21 21:15:02,948 [INFO] 172.18.0.7 - - [21/Dec/2025 21:15:02] "GET /info/1 HTTP/1.0" 200 -
```

GET http://localhost/info/1 Send

Docs Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers (5) Test Results 200 OK 9 ms 281 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "price": 40.0,
4   "quantity": 5,
5   "title": "How to get a good grade in DOS in 40 minutes a day",
6   "topic": "distributed systems"
7 }
```

```
frontend | 2025-12-21 21:14:21,497 [INFO] WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
frontend | * Running on all addresses (0.0.0.0)
frontend | * Running on http://127.0.0.1:5000
frontend | * Running on http://172.18.0.6:5000
frontend | 2025-12-21 21:14:21,497 [INFO] Press CTRL+C to quit
frontend | 2025-12-21 21:15:02,937 [INFO] CACHE MISS info:1
frontend | 2025-12-21 21:15:02,948 [INFO] 172.18.0.7 - - [21/Dec/2025 21:15:02] "GET /info/1 HTTP/1.0" 200 -
frontend | 2025-12-21 21:15:45,298 [INFO] CACHE HIT info:1
frontend | 2025-12-21 21:15:45,298 [INFO] 172.18.0.7 - - [21/Dec/2025 21:15:45] "GET /info/1 HTTP/1.0" 200 -
```

GET

http://localhost/search/undergraduate

Send

Docs

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

↺

200 OK

• 20 ms

• 331 B

🌐

📄 Save Response

...

{ } JSON

▶ Preview

🖼 Visualize

▼

🔍

📄

🔗

```
1  [  
2    {  
3      "id": 3,  
4      "title": "Xen and the Art of Surviving Undergraduate School"  
5    },  
6    {  
7      "id": 4,  
8      "title": "Cooking for the Impatient Undergrad"  
9    },  
10   {  
11     "id": 6,  
12     "title": "Why theory classes are so hard"  
13   }  
14 ]
```

GET

http://localhost/search/undergraduate

Send

Docs

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

↺

200 OK

• 10 ms

• 331 B

🌐

📄 Save Response

...

{ } JSON

▶ Preview

🖼 Visualize

▼

🔍

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🔗

```
1  [  
2    {  
3      "id": 3,  
4      "title": "Xen and the Art of Surviving Undergraduate School"  
5    },  
6    {  
7      "id": 4,  
8      "title": "Cooking for the Impatient Undergrad"  
9    },  
10   {  
11     "id": 6,  
12     "title": "Why theory classes are so hard"  
13   }  
14 ]
```

POST http://localhost/purchase/1 Send

Docs Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (5) Test Results 200 OK 80 ms 349 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "item_id": 1,
3   "order_id": 1,
4   "qty": 1,
5   "replicated_to_peer": true,
6   "status": "success",
7   "timestamp": "2025-12-21 21:16:43",
8   "title": "How to get a good grade in DOS in 40 minutes a day",
9   "total_price": 40.0
10 }
```

After purchase

GET http://localhost/info/1 Send

Docs Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (5) Test Results 200 OK 14 ms 281 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "price": 40.0,
4   "quantity": 4,
5   "title": "How to get a good grade in DOS in 40 minutes a day",
6   "topic": "distributed systems"
7 }
```

```
frontend | 2025-12-21 21:17:26,910 [INFO] CACHE MISS info:1
frontend | 2025-12-21 21:17:26,916 [INFO] 172.18.0.7 - - [21/Dec/2025 21:17:26] "GET /info/1 HTTP/1.0" 200 -
```

Before updating the catalog during a purchase request, the backend sends an invalidate request to the front-end.
This removes the cached entry to ensure strong consistency.

```
frontend | 2025-12-22 17:23:50,639 [INFO] Press CTRL+C to quit
frontend | 2025-12-22 17:25:30,119 [INFO] CACHE MISS info:1
frontend | 2025-12-22 17:25:30,130 [INFO] 172.18.0.7 - - [22/Dec/2025 17:25:30] "GET /info/1 HTTP/1.0" 200 -
frontend | 2025-12-22 17:25:33,762 [INFO] CACHE HIT info:1
frontend | 2025-12-22 17:25:33,762 [INFO] 172.18.0.7 - - [22/Dec/2025 17:25:33] "GET /info/1 HTTP/1.0" 200 -
frontend | 2025-12-22 17:26:08,026 [INFO] 172.18.0.5 - - [22/Dec/2025 17:26:08] "POST /invalidate HTTP/1.1" 200 -
frontend | 2025-12-22 17:26:08,036 [INFO] 172.18.0.3 - - [22/Dec/2025 17:26:08] "POST /invalidate HTTP/1.1" 200 -
frontend | 2025-12-22 17:26:08,063 [INFO] 172.18.0.1 - - [22/Dec/2025 17:26:08] "POST /purchase/1 HTTP/1.1" 200 -
frontend | 2025-12-22 17:28:11,769 [INFO] CACHE MISS info:1
frontend | 2025-12-22 17:28:11,775 [INFO] 172.18.0.7 - - [22/Dec/2025 17:28:11] "GET /info/1 HTTP/1.0" 200 -
```

Scenario	Request Type	Cache Enabled	Average Response Time (ms)	Cache Status
1	info(1)	No	56	CACHE MISS
2	info(1)	No	19	CACHE MISS
3	info(1)	Yes	24	CACHE MISS
4	info(1)	Yes	9	CACHE HIT
5	info(1)	Yes (after purchase)	14	CACHE MISS (after invalidate)

The results show that cache hits significantly reduce response time compared to cache misses.
After a write operation, cache invalidation introduces additional overhead, but ensures strong consistency

	catalog1 ● bazar-online-book-s 5001:5001 ↗		
	catalog2 ● bazar-online-book-s		
	order1 ● bazar-online-book-s 5002:5002 ↗		
	order2 ● bazar-online-book-s		
	frontend ● bazar-online-book-s 5000:5000 ↗		
	nginx ● bazar-online-book-s 80:80 ↗		

Docker containers